allocation that exaggerated the risk of the 4 percent rule, I used a rather generous and unrealistic “perfect foresight assumption.” For each country and in each retirement year, I used the asset allocation for local market investors (between stocks, bonds, and bills) that supported the highest withdrawal rate. In many cases, though not all, this meant using 100 percent stocks.

Using the updated data set through 2015 and the perfect foresight assumption, the calculated SAFEMAX meets or exceeds 4 percent in Sweden (4.5 percent), Canada (4.4 percent), New Zealand (4.1 percent), Denmark (4.1 percent), and the United States (4 percent)—only five of the twenty countries. In seven countries, the perfect foresight assumption yields a SAFEMAX below 2 percent for hypothetical retirees investing in their local markets.

The perfect foresight assumption is quite unrealistic, though. As mentioned, the optimal asset allocation is often 100 percent stocks, which is too aggressive for most retirees, and of course no one knows the best asset allocation in advance. For this reason, it is worthwhile to focus on a more plausible 50/50 retirement asset allocation. In terms of the SAFEMAX, 50 percent stocks and 50 percent bills generally outperforms 50 percent stocks and 50 percent bonds in the data set. For bonds, the additional returns over bills failed to compensate for their additional volatility in a retirement income plan, so I will further consider results for the stocks and bills case. Bengen used intermediate-term government bonds—which seem to hold a sweet spot in the trade-off between returns and volatility—to yield the highest sustainable spending. But this bond option is not available in the Global Returns Dataset. For what is available, bills work better than long-term bonds for the aforementioned reason that their lower volatility helps more than the lower return hurts. The results for the 50/50 asset allocation can be seen in Exhibit 5.2.

Exhibit 5.2
Maximum Sustainable Withdrawal Rates for Retirees
Global Returns Dataset, 1900–2015
Asset Allocation: 50% Stocks and 50% Bills

				Withdrawal Rate = 4%		Withdrawal Rate = 5%	
	SAFEMAX	SAFEMAX Year	10th Percentile	# Years in Worst Case	% Failures within 30 Years	# Years in Worst Case	% Failures within 30 Years